REIT FFO Comparability Pipeline

Project Plan

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1 Project Summary

Funds from operations (FFO) is the main earnings measure for REITs, but the company-defined versions (Core FFO, Normalized FFO, CFFO, FFO as Adjusted, AFFO) are not comparable across peers. Each REIT adds back different items, companies change their own definitions over time, and the reconciliations sit in quarterly filings with different layouts.

One-sentence description. Our application helps REIT equity analysts and real estate investors make apples-to-apples valuation and peer-comparison decisions across all U.S. publicly traded multifamily REITs by using an AI agent that monitors SEC EDGAR daily to track which REITs are listed, collect every quarterly earnings filing since 2018, and extract each FFO reconciliation, to produce a continuously updated, restated FFO history on one common definition, with source-page citations and plain-language explanations of how each REIT's reported metrics differ.

The pipeline is written in Python on the Claude API and:

1. **tracks the universe**: finds every listed multifamily REIT, including those listed at any point since 2018 that have since been acquired, merged or liquidated;
2. **collects** each quarterly earnings document from SEC EDGAR;
3. **extracts** every line of each FFO reconciliation into structured data;
4. **restates** every company-quarter to one fixed common definition, applied retroactively (the arithmetic runs in code, not in the AI);
5. **explains** how each REIT's reported metrics differ, with page citations; and
6. **updates itself daily**, picking up new quarters, new REITs and exits.

Track universe → **Collect** (EDGAR API) → **Extract** (AI) → **Restate** (Python) → **Explain and report**

runs daily on a schedule

2 Scope

- **Companies:** every U.S. exchange-listed REIT that meets the multifamily rule (Phase 4), at any point from 2018 to today.
- **Period:** Q1 2018 to the latest reported quarter, then every new quarter as it is filed. Roughly 35 quarters of history.
- **Metrics:** NAREIT FFO, each company's own "core" metric, and AFFO.
- **Common definition:** one fixed definition, today's, applied to every quarter regardless of what definition the company used at the time.

- **Disclosure limits:** where an older filing combines or omits an item the definition needs, the pipeline searches the 10-Q notes; if the item still cannot be separated, the quarter is flagged "restated with limits" rather than guessed.

Current universe (October 2026)

Ticker	Company	Markets	"Core" metric / note
VMRK	Vivmark Residential	Coastal	Formed Aug. 2026 from AvalonBay (Core FFO) and Equity Residential (Normalized FFO); history kept as two companies before the merger
ESS	Essex Property Trust	West Coast	Core FFO
UDR	UDR	Mixed	FFO as Adjusted
MAA	Mid-America Apartment Communities	Sunbelt	Core FFO, Core AFFO
CPT	Camden Property Trust	Sunbelt	Core FFO, Core AFFO
IRT	Independence Realty Trust	Sunbelt/Midwest	CFFO; acquiring Centerspace
NXRT	NexPoint Residential Trust	Sunbelt	Core FFO
BRT	BRT Apartments	Sunbelt	No core metric (FFO, AFFO)
CSR	Centerspace	Midwest/Mountain	Core FFO; being acquired by IRT
CLPR	Clipper Realty	New York City	AFFO; inclusion decided by the multifamily rule

Former REITs in the 2018+ history (examples)

Veris Residential (taken private 2026), Elme Communities (liquidated 2026), Aimco (liquidating), AIR Communities (taken private 2024), Preferred Apartment Communities and Bluerock Residential (acquired 2022). Including them avoids **survivorship bias**: a history built only from today's survivors would skew every comparison.

3 Testing Strategy

The hand-built set is the answer key the pipeline is graded against:

- 15 supplementals collected by hand: MAA, CPT, IRT, NXRT and BRT for Q4 2025–Q2 2026.
- MAA reconciled by hand for all three quarters (Excel workbook).
- Spot checks: one quarter each for three or four other REITs, plus at least two pre-2020 quarters to test older layouts.

4 Project Steps

4.1 Phase 1 — Scope and Background

☐ Confirm scope: all U.S.-listed multifamily REITs, Q1 2018 to present. *Done.*

☐ Read the Nareit FFO White Paper and list the NAREIT-standard adjustments.

4.2 Phase 2 — Answer Key Documents

☐ Collect 15 supplementals by hand. *Done.*

☐ Rename files consistently and store in `data/raw/`. *Done.*

☐ Replace the BRT Q4 2025 file with a text-based copy from EDGAR.

☐ Record the EDGAR filing each file came from (form, date, exhibit).

4.3 Phase 3 — Manual Baseline (MAA)

☐ Q2 2026 reconciliation. *Done (worked example).*

☐ Q4 2025 and Q1 2026 reconciliations; all checks show OK.

☐ Fill in the Adjustment Log for all three quarters.

4.4 Phase 4 — Definitions

☐ Build the adjustment map across the five answer-key REITs.

☐ Write the common FFO definition: which items are included and why.

☐ Write the **multifamily rule** (e.g. more than half of NOI or real estate from apartments) used to decide which REITs are in the universe.

☐ Define the "restated with limits" flag and when it applies.

4.5 Phase 5 — Collection Agent (fixed list, current quarters)

☐ Set up Python, the Claude API key and the repository. *Repository done.*

☐ Write EDGAR tools: list a company's filings, list a filing's exhibits, download a file. Send the User-Agent header the SEC requires.

☐ Agent finds each earnings 8-K and picks the exhibit holding the FFO reconciliation.

☐ Checks: file mentions "Funds from Operations" and the right period end.

☐ Run on the five answer-key REITs; compare against the 15 hand-collected files.

4.6 Phase 6 — Extraction and Restatement

☐ Define the JSON schema for one reconciliation (lines, amounts, per-share, pages).

☐ Prompt Claude to fill it and to flag unclear items instead of guessing.

☐ Check in code that extracted lines sum to reported totals.

☐ Map line items to common categories; compute restated FFO in Python.

- [ ] Generate a plain-language note per company-quarter.
- [ ] Test on MAA against the manual baseline until it matches.

4.7 Phase 7 — Accuracy Testing

- [ ] Collection: correct filing and exhibit for every answer-key company-quarter.
- [ ] Extraction: match to $0.01 per share.
- [ ] Error log by type: wrong document, wrong number, misclassified adjustment, wrong page, missed item.

4.8 Phase 8 — Live Universe and Daily Updates

- [ ] Pull all REITs (SIC code 6798) with tickers from the SEC company list.
- [ ] Claude classifies each candidate against the multifamily rule, citing the 10-K.
- [ ] New entries are proposed for review before they join the universe.
- [ ] Detect exits from Form 25 (delisting) and Form 15 (deregistration); keep history.
- [ ] Schedule a daily run with GitHub Actions: new quarters, new REITs, exits.

4.9 Phase 9 — Backfill to Q1 2018

- [ ] Identify every multifamily REIT listed at any point since 2018, including former ones.
- [ ] Collect and extract all quarters from Q1 2018 onward.
- [ ] Search 10-Q notes where reconciliations lack detail; flag what cannot be separated.
- [ ] Spot-check pre-2020 quarters by hand.
- [ ] Report the share of quarters fully restated vs. restated with limits.

4.10 Phase 10 — Deliverables

- [ ] Full proposal after fall break (scope, method, universe, test plan).
- [ ] Code repository on GitHub with a README.
- [ ] Restated FFO history (Excel and data files).
- [ ] Interactive FFO dashboard (published on Vercel), as specified in Section 5.
- [ ] Write-up of findings and accuracy results.
- [ ] Short resume summary of the project.

5 Final Product: FFO Dashboard

A web dashboard, rebuilt automatically after each daily pipeline run, with:

1. **REIT selector.** A drop-down menu listing every REIT in the universe (current and former), plus a quarter selector (Q1 2018 to latest).
2. **Restated FFO on the common definition.** The headline figure (total and per share) using this project's calculation, shown beside the company's own reported FFO and "core" metric.
3. **The formula.** The calculation written out step by step, e.g.

$$\text{FFO} = \text{Net income} + \text{RE depreciation} - \text{Gains on sale} + \text{JV depreciation} + \text{NCI}$$

$$\text{Common FFO} = \text{FFO} \pm \text{adjustments under the common definition}$$

   with the REIT's actual numbers substituted into each term.
4. **Line-item references.** Every term in the formula shows the amount, the exact line-item name as printed in the filing, and its location, e.g. *"Legal costs, settlements and (recoveries), net" — $53,000K — page 10*, with a link to the source document on SEC EDGAR.
5. **Flags.** Quarters "restated with limits" are marked, with the item that could not be separated.

How the data supports it. The extraction step stores, for every number, the line-item label as printed, the page and the source filing. The restatement step records which items the common definition kept or removed. The dashboard only displays these stored records; it does no calculation of its own, so every figure on screen traces back to a page in a filing.

6 Data Sources

- SEC EDGAR via the free API (`data.sec.gov`): 8-K earnings exhibits, 10-Q and 10-K filings, Forms 25 and 15, and the company list with SIC codes.
- REIT investor relations sites: quarterly supplementals (answer key).
- Nareit FFO White Paper (baseline definition).
- Claude API: classification, extraction and explanations.

References

AvalonBay–Equity Residential merger closes as Vivmark

IRT to acquire Centerspace

Veris Residential taken private

Elme Communities NYSE trading suspended

Aimco liquidation update

MAA Q2 2026 8-K